ITERATING THE CONSTRUCTION PRODUCES A FAITHFULLY FLAT IND-ETALE ALGEBRA OVER WHICH EVERY FAITHFULLY FLAT ETALE MAP HAS A RETRACTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the construction of the colimit of the iterates with its faithful flatness and ind-etale properties, the descent of an etale map over the colimit to a stage with the verification of faithful flatness, and the construction of the retraction from the defining property of the iteration.

Editorial note. The label lemma-first-construction names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-first-construction –start 1574 –end 1598 –local-id 28001 returned RESULT=OK (L28001 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The one step construction and its universal property. Same chapter, the construction T(A) of the first step together with the retraction property it enjoys. Supplies the iteration T^n and the algebra maps B' -> T^{n+1}(A) used at the end.
- Faithful flatness and ind-etale-ness of a filtered colimit. Algebra, the lemma that a filtered colimit of faithfully flat algebras is faithfully flat over each stage, and Same chapter, the lemma that an iteration of ind-etale maps is ind-etale. Supply the faithful flatness of C over every T^n(A) and the ind-etale property of C over A.
- Descent of an etale algebra along a colimit. Algebra, the lemma describing an etale algebra over a filtered colimit as the base change of an etale algebra over some stage. Supplies the index n and the etale ring map T^n(A) -> B' with B = C tensor over T^n(A) with B'.

Source reference. Original lemma 090N: sources/stacks-chow/tags.txt, source lines 11416–11416. Prerequisite pointer invoked at proof line(s) 1587; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:proetale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00U2: sources/stacks-chow/tags.txt, source lines 1096–1096. Prerequisite pointer invoked at proof line(s) 1591; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 097L: sources/stacks-proetale/pinned-original-source.tex, source lines 1408–1408. Prerequisite pointer invoked at proof line(s) 1588; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Given a ring A there exists a faithfully flat ind-étale A-algebra C such that every faithfully flat étale ring map $C \to B$ has a retraction.*

Proof. Set $T^1(A) = T(A)$ and $T^{n+1}(A) = T(T^n(A))$. Let

$$C = \operatorname{colim} T^n(A)$$

This algebra is faithfully flat over each $T^n(A)$ and in particular over A, see Algebra, Lemma 090N. Moreover, C is ind-étale over A by Lemma 097L. If $C \to B$ is étale, then there exists an n and an étale ring map $T^n(A) \to B'$ such that $B = C \otimes_{T^n(A)} B'$, see Algebra, Lemma 00U2. If $C \to B$ is faithfully flat, then $\operatorname{Spec}(B) \to \operatorname{Spec}(C) \to \operatorname{Spec}(T^n(A))$ is surjective, hence $\operatorname{Spec}(B') \to \operatorname{Spec}(T^n(A))$ is surjective. In other words, $T^n(A) \to B'$ is faithfully flat. By our construction, there is a $T^n(A)$-algebra map $B' \to T^{n+1}(A)$. This induces a C-algebra map $B \to C$ which finishes the proof. $\square$